

ReNikud: Audio-Supervised Hebrew Grapheme-to-Phoneme Conversion

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Abstract—Grapheme-to-phoneme (G2P) conversion for Modern Hebrew is needed for applications like text-to-speech (TTS), but is challenging due to the language’s abjad writing system, which leaves vowels largely unwritten, creating widespread ambiguity. Standard approaches first predict vowel diacritics (nikud) to produce International Phonetic Alphabet (IPA) transcriptions, but this is limited: vocalization data is scarce and laborious to produce, it does not specify features such as lexical stress, and it reflects formal grammatical rules rather than everyday spoken pronunciation. Direct sequence-to-sequence IPA prediction, meanwhile, struggles on limited data and produces frequent hallucinations due to a lack of character-level alignment. Our method, ReNikud, overcomes these limitations with two key insights: (1) Weak audio supervision via a phoneme-based automatic speech recognition (ASR) pseudo-labeling pipeline on thousands of hours of unlabeled Hebrew audio, yielding phonemic transcriptions that reflect natural spoken norms without manual annotation. (2) A pseudo-vocalization architecture that predicts IPA phonemes at each character position, enforcing character-level alignment to mitigate hallucinations. Results on existing Hebrew G2P benchmarks and new targeted test suites for spoken Hebrew show that ReNikud surpasses previous state-of-the-art methods. We will release our code and trained models to support further work on Hebrew TTS and speech technologies.

Index Terms—Grapheme-to-Phoneme, Text-to-Speech, Modern Hebrew, Weakly Supervised Learning, Automatic Speech Recognition, Lexical Stress

I. INTRODUCTION

With increasing interest in text-to-speech (TTS) systems for low-resource languages, the modern Hebrew language poses particular challenges. Hebrew is written as an *abjad*—a writing system that normally does not indicate vowel sounds [1]. Consequently, many words are homographs and require semantic context for correct pronunciation. For example, the word מספרים can be read as /mispakʰim/¹ (*numbers*), /mispakʰajim/ (*scissors*), or /mesapʰim/ (*telling*). As this ambiguity confounds TTS generations, leading open-source approaches first predict *nikud* (vocalization, i.e., conventional vowel diacritics) to condition synthesis [2], [3].

However, there are substantial gaps between vocalization and fully-specified International Phonetic Alphabet (IPA) transcriptions for grapheme-to-phoneme (G2P) conversion.

One issue is the mismatch between written and spoken Hebrew. Firstly, nikud does not fully specify phonetic features such as lexical stress, e.g., בירה may be pronounced as /bʰiʁa/

(*beer*) or /bʰisʰa/ (*capital city*). Secondly, nikud conventionally reflects traditional grammatical rules that do not match everyday spoken Hebrew. For instance, the word ובדרך is formally vocalized and pronounced as /ubadʰeʁeʁ/, but in natural speech, it is pronounced /vebadʰeʁeʁ/. Similarly, בירושלים is formally /bʰiʁuʃalʰajim/, but native speakers typically say /beʁeʁuʃalʰajim/ [4], [5].

Another issue is a lack of scalable data sources for directly learning G2P. Vocalization models [6], [7] learn from Hebrew text with manually annotated vowel diacritics, data which is scarce and laborious to produce. Similarly, methods such as [3], [8] that learn from IPA annotations are bottlenecked by data availability. Conversely, another abundant source of data on pronunciation exists—unlabelled Hebrew audio—but existing methods can only learn from text.

Finally, while methods using vocalization poorly reflect spoken pronunciation, they have been shown to outperform direct sequence-to-sequence (seq2seq) prediction of IPA [3], the latter of which struggles to learn on limited data and produces frequent hallucinations due to a lack of enforced character-level alignment with the input transcript.

Our method, *ReNikud*², directly addresses these challenges. To reflect spoken norms and unlock abundant training data, we propose a novel pipeline to train G2P with *weak supervision from unlabeled Hebrew audio*. By applying an automatic speech recognition (ASR)-based pseudo-labeling pipeline, we are able to extract pronunciation information and train on thousands of hours of recordings, a scalable approach that effectively learns spoken Hebrew norms. To enforce character-level alignment between Hebrew characters and phonemes, we introduce a *pseudo-vocalization* architecture. Like traditional vocalization, this predicts phonetic content aligned with each character position, but rather than predicting traditional *nikud* symbols it directly predicts character-aligned IPA phonemes. By using the orthographic structure of Hebrew text, this mitigates hallucinations and increases data efficiency relative to seq2seq baselines, as we demonstrate.

Our results on existing Hebrew G2P benchmarks show that ReNikud succeeds in leveraging audio data and better reflecting Hebrew speech. In addition, we propose additional test suites and targeted evaluations to show improvements on difficult phonetic cases reflecting spoken language.

*Equal contribution

¹We place the IPA stress mark directly before the stressed vowel, following TTS conventions.

²Rethinking *Nikud*

We will release our code, data and trained models to spur work on Hebrew speech technologies.

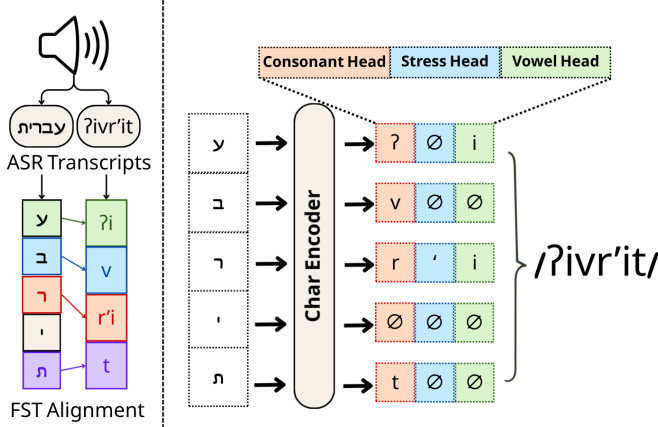


Figure 1: **System overview.** We first pseudo-label audio (left) by creating a many-to-one FST alignment between unvocalized Hebrew text and IPA phonemes derived from two parallel ASR runs applied to Hebrew audio. We then train a pseudo-vocalization architecture (right) where unvocalized Hebrew characters are passed through a character encoder to predict a phonetic triplet (consonant, stress, and vowel) at each position via parallel classification heads.

II. METHOD

The ReNikud pipeline consists of two stages, shown in Figure 1: audio pseudo-labeling (Section II-A), and our pseudo-vocalization architecture trained on this data (Section II-B).

A. Audio Pseudo-Labeling

To learn pronunciation from unlabeled audio at scale, we construct a pipeline that extracts character-aligned IPA annotations using two parallel ASR systems, as shown in Figure 1 (upper left). We extract Hebrew orthographic transcripts with a standard pretrained Hebrew ASR model, and IPA transcripts with a custom ASR model trained to output IPA when applied to Hebrew audio. By applying both ASR systems to a large-scale, unlabeled Hebrew audio corpus, we extract parallel Hebrew text and IPA transcriptions that serve as *pseudo-labels*, providing weak supervision for our downstream G2P model.

Importantly, we retain only pairs where the Hebrew and IPA transcripts agree on the number of words and where every word passes the FST alignment procedure described below. This filtering retains approximately [TODO: XXX%] of the original samples. While strict, this selection criterion still yields several million [Morris: make precise?] aligned sentence from the large dataset, selecting for more reliably transcribed utterances.

Finally, we perform a string alignment process based on the Hebrew orthography’s abjad structure. In general, abjads encode consonants linearly with interleaved, unwritten vowels, meaning that graphemes map monotonically to (consonant, vowel) pairs. As Hebrew also has unwritten lexical stress, each

grapheme maps to a *phonetic triplet* encoding such a pair and a binary stress indicator. We find this alignment with a simple finite state transducer (FST) handling known consonant values, including one-to-many (e.g., כ→/b, v/) and many-to-one (e.g., ו, ת→/t/) mappings, as well as orthographic complexities such as:

- **Digraphs:** Loanwords in Modern Hebrew frequently use an apostrophe (*geresh*) to denote non-native phonemes (e.g., ׳ for /z/; the base letter ז normally represents /z/). The FST assigns the digraph phoneme to the base letter and passes over the *geresh*.
- **Word-final ח (/χ/)** may occur with an additional *preceding* /a/ vowel (*patah gnuva*), e.g., לוח /l'uaχ/. We handle this reordering with a special combined /aχ/ value for the vowel slot.
- **Silent letters:** The Hebrew letters ו, י, ה, א may either indicate consonant sounds or may be silent (*matres lectionis*). We handle the latter case with a null (∅) consonant class.

An example of the resulting alignment between Hebrew characters and phonetic triplets is shown in Table I.

Word	Char	Consonant	Vowel	Stress
שלום (/ʃal'om/)	ש	/ʃ/	/a/	0
	ל	/l/	/o/	1
	ו	∅	∅	0
	ם	/m/	∅	0
צִיפּוֹס (/tʃ'ips/)	צ	/tʃ/	/i/	1
	,	∅	∅	0
	י	∅	∅	0
	פ	/p/	∅	0
	ס	/s/	∅	0
תפוח (/tap'uaχ/)	ת	/t/	/a/	0
	פ	/p/	/u/	1
	ו	∅	∅	0
	ח	∅	/aχ/	0

Table I: **Examples of FST-derived alignment** between Hebrew characters and phonetic triplets (consonant, vowel, stress).

B. Pseudo-Vocalization Architecture

Our goal is to create a Hebrew G2P model that maps unvocalized Hebrew text directly to IPA strings. Because Hebrew is an *abjad*, there is a strong, local relationship between individual written letters and their phonetic realizations. Standard sequence-to-sequence models fail to exploit these localized properties, leading to hallucinations.

To mitigate this, we frame the G2P task as a constrained, per-character classification problem—a method we term *Pseudo-Vocalization*, illustrated in Figure 1 (right). While a single Hebrew character typically corresponds to more than one IPA symbol (e.g., a consonant followed by a vowel), we resolve this by having every Hebrew letter independently predict exactly one phonetic triplet, as described in Section II-A.

The core model is a character-level transformer encoder. On top of this encoder, we attach three parallel, independent

classification heads that simultaneously predict the phonetic attributes for each character directly from the encoder’s hidden states:

- **Consonant Head:** Selects from 25 IPA consonants or null ($\emptyset$).
- **Vowel Head:** Selects from 5 vowels, null ($\emptyset$), or the special /a χ / token (see Section II-A).
- **Stress Head:** Binary classifier for lexical stress.

At inference time, realizations are predicted by taking the argmax of logits for each head. In addition, we apply *constrained decoding* to enforce hard constraints on Hebrew letters and phonetic realizations: the argmax is calculated only over possible consonantal realizations of a letter (e.g., $\aleph$ can only be realized as /b/ or /v/). We also enforce a word-level constraint that exactly one lexical stress is predicted.

C. Renikud Classifier

The core model is built upon dicta-il/dictabert-large-char, a character-level BERT encoder pre-trained on Hebrew. We selected this base encoder because our method requires character-level representation. On top of this base encoder, we attach three independent classification heads that simultaneously predict the following phonetic elements for every input character:

- **Letter Head:** Predicts the base consonant from a predefined set of 25 possible classes.
- **Vowel Head:** Predicts the associated vowel from a set of 7 possible classes.
- **Stress Head:** A binary classifier indicating whether the character carries primary lexical stress (true/false).

III. EXPERIMENTS AND RESULTS

A. Experimental Setup

For our base ASR model, we employ the Whisper Large v3 Turbo checkpoint adapted for Hebrew by ivrit.ai, which we subsequently fine-tune on text-and-IPA objective. Our training process is divided into two distinct stages:

Dataset	Size	Role
SASpeech [9]	~18h	Pretraining (Stage 1)
Recital	~46h	Pretraining (Stage 1)
ILSpeech [3]	~2h	Fine-tuning (Stage 2)

Table II: **Data sources** for IPA ASR training.

Stage 1: Pretraining. During the initial pre-training phase, we utilized two primary corpora. The first is a subset of the *SASpeech* dataset [9]. While the full corpus contains roughly 30 hours of audio, the automatic audio files that contain 26 hours we filtered short files with single word, we used the ivrit’s ASR to filter all the audio with a lot of “umm..” by doing this we trim it down to approximately 14 hours of the automatically aligned data. with the 4 manual data we got about 18 hours. The second is the crowd-sourced *Recital* dataset, consisting of approximately 46 hours of raw audio. we processed the datasets’ existing Hebrew transcripts directly through Phonikud [3] to produce IPA.

Stage 2: Fine-tuning and Evaluation of the ASR Following pre-training, the model was fine-tuned on the manual ipa *ILSpeech* dataset [3], which comprises approximately 2 hours of audio. From the dataset, we extracted exactly 150 recordings for our held-out test set, reserving the remainder for fine-tuning.

To assess the ASR-IPA capabilities, we evaluated the fine-tuned model on two datasets: the held-out test set of 150 manually annotated sentences from *ILSpeech*, and *Michel*, a synthetic out-of-domain dataset comprising 250 sentences. We report Character Error Rate (CER), Word Error Rate (WER), Vowel Error Rate (VER), and Stress Error Rate (SER). Results are detailed in Table III.

Test Set	CER	WER	VER	SER
ILSpeech (150 seq.)	0.0240	0.1120	0.0224	0.9240
Michel (250 seq.)	0.0478	0.2216	0.0321	0.1767

Table III: **ASR IPA evaluation results** on the in-domain ILSpeech and out-of-domain synthetic Michel datasets.

[Max: is 350 sentences enough for showing the results?]
[Max: is there a way to explain that michel is a legit dataset?]

Stage 3: Training of Renikud For our primary large-scale dataset, we utilized Knesset Vox [10], specifically drawing upon approximately 1.7k hours of parliamentary recordings from its training set. Knesset Vox has ready to use transcripts by using ASR and forced alignment between the already made scripts from the knesset parliament transcripts. Because the Whisper ASR model experiences performance degradation on longer audio inputs (exceeding 25 seconds), we bypassed the original Knesset Vox sentence annotations and re-segmented the raw audio into optimized 5–15-second clips to ensure correct results of phonetic transcriptions with the ASR. We then processed these shorter segments using both a standard Hebrew Whisper model and our custom IPA-tuned Whisper model to extract parallel Hebrew text and IPA transcriptions. After extracting the pseudo-labels and applying the FST alignment process described in Section II-A, To ensure memory stability and prevent out-of-memory (OOM) errors during training, we filtered out sentence-length outliers using the standard $1.5 \times \text{IQR}$ rule. Specifically, we discarded sequences with lengths falling outside the bounds of $[Q_1 - 1.5 \times \text{IQR}, Q_3 + 1.5 \times \text{IQR}]$. Ultimately, this weak supervision pipeline yielded a dataset of approximately 1.52 million aligned Hebrew-to-IPA sentences. aligned Hebrew-to-IPA sentences. [Morris: does Knesset Vox have gold transcripts? do we ignore them?] [Max: Yes, but we create shorter clips of 5-15 and it will be a headache to use the gold labels also probably were made with whisper] [Morris: should be explained concisely][Max: added some explanation about the transcriptions]

Stage 4: Evaluation of Renikud To benchmark our proposed ReNikud architecture, we evaluated its grapheme-to-phoneme mapping capabilities against the standard Phonikud baseline.

Cat.	Input	Target	IPA	Cat.	Input	Target	IPA
Gender	תגיד, הכל בסדר איתך?	איתך	/ʔit'aχ/	Names	ליהי איפה החולצה הכחולה שלי?	ליהי	/li'hi/
Acronyms	הוא שירת באמ"ן	באמ"ן	/be'am'an/	Homogr.	הוא קרא ספר מעניין בזמן שהוא ישב אצל ספר נשים.	ספר	/s'efek/-/sap'ak/
Penult.	קניתי לחם טרי	לחם	/l'eχem/	Min. Str.	היה לו נחת רוח מההישגים של ילדיו / המטוס נחת בשלום על המסלול	נחת	/n'aχat/-/naχ'at/
Rare Ph.	הבאתי ציפס בשבילך, תאכל מהר.	ציפס	/tʃ'ips/	Slang	הוא בן אדם גנוב לגמרי.	גנוב	/gan'uv/
Foreign	איזה וויב טוב יש פה	וויב	/v'ajb/				

Table IV: Selected examples from BITUI, split by category (Cat.). Abbreviated category names are penultimate stress (Penult.), rare phonemes (Rare Ph.), homographs (Homogr.), and minimal stress pairs (Min. Str.).

B. BITUI Testset

BITUI³ testset consists of a targeted evaluation corpus designed to systematically assess a model's capacity for complex linguistic disambiguation in context-dependent orthographies. BITUI testset except of the ilspeech, the generation of the sentences was created using Gemini-3.1-pro and then we fixed it manually. the high result of the gemini in the graphs might be the results of bias towards his sentences. ILspeech was updated from Phonikud [3] mistakes were fixed like e.g. ותוכן, it was אבטרים [Morris: how many samples are in BITUI?]

Targeted Evaluation Methodology:

The dataset is strictly categorized into five distinct linguistic challenges:

- **Morphological and Syntactic Ambiguity (gender):** Hebrew pronouns, verbs, and adjectives frequently share identical orthography but shift in pronunciation based on the gender of the subject or object.
 - Example: אני נותן לך את הספר תקרא אותו
 - Targeted Label: Index 2 (לך) → /leχ'a/ (Resolved via the masculine future verb תקרא).
- **Semantic Ambiguity (homographs):** This category tests identically spelled words with completely divergent meanings and phonetic structures.
 - Example 1: חתכתי בצל למרק (I cut an **onion** for the soup) → /bats'al/.
 - Example 2: ישבנו בצל העץ (We sat **in the shade** of the tree) → /bets'el/. [Max: not sure about this example]
- **Colloquialisms and Slang (slang):** Modern spoken Hebrew frequently diverges from prescriptive dictionary rules.
 - Example: איזו פאדיחה הייתה אתמול
 - Targeted Label: פאדיחה → /fad'ixa/
- **Foreign Loanwords and Rare Phonemes (loanwords):** Hebrew orthography adapts to non-native phonemes (e.g., /w/, /d/, /t/) using modifying characters (geresh) or unusual letter combinations.
 - Example: הזמנתי המבורגר וציפס
 - Targeted Label: ציפס → /tʃ'ips/

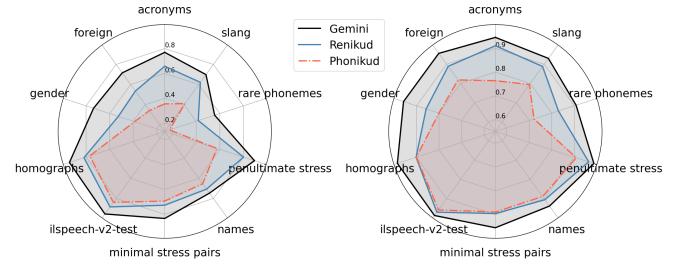


Figure 2: Word accuracy rate (left) and character accuracy rate (right). [Max: should i add the name Gemini-3.1-pro? in the graph or in caption?] [Morris: in caption or paper text is fine. It should be a gray dotted line, not solid black, and in caption we should say that it provides an upper bound]

- **Baseline Control (ilspeech-v2-test):** Main testset of phonikud and also to test general speech classification correction.

– Example: הייתה מעורבת ממילא באינספור פרויקטים צבאיים מתקדמים

Evaluation Methodology: Predictions were compared against human-annotated gold-standard IPA transcriptions. Performance is reported via Word Error Rate (WER) and Character Error Rate (CER). The overall model performance (reported as OVERALL) is calculated as a micro-average of the error rates across all distinct evaluation categories, encompassing both the challenge suite and the generalized test set in [3] as seen in

Results and Analysis: The comparative results are detailed in Table V. For example, ReNikud accurately predicts the colloquial slang פאדיחה as /fad'ixa/, whereas the Phonikud incorrectly outputs the /padiχ'a/. Furthermore, ReNikud successfully maps non-native phonemes in foreign loanwords, correctly resolving the English 'w' in ווינר (/w'ineχ/) and the 'j' sound in גוב (/hadz'ob/), which the Phonikud misclassifies as (/vin'eχ/ and /hadz'ov/). Gemini-3.1-pro results used as an upper bound for the possible results.

[Morris: do we have a category for things like vebaderex which we discussed in the intro?] [Max: nope]

[Morris: add something mentioning that ILSpeech test set is expanded/updated (so it's clear why Gemini results aren't identical to the ones in Phonikud paper)]

[Morris: add qualitative results – still missing full results]

³Benchmark for Individual-word Testing of Underspecified IPA

	Gender	Acronyms	Penult. Stress	Rare Phonemes	Foreign	Names	Stress Homog.	Slang	IL-Speech-test	OVERALL
ReNikud (Ours)	49/17	34/9	20/4	59/18	46/11	29/9	32/21	38/11	16/4	27/8
ReNikud (Phonikud Data)	47/18	76/26	30/6	82/30	47/13	33/11	34/21	44/15	19/5	33/11
Phonikud (Baseline)	62/22	64/24	42/9	83/28	66/18	35/11	36/21	59/21	20/5	36/12

Table V: Detailed evaluation on the Hebrew G2P challenge suite (WER/CER, in %). Our proposed ReNikud model trained on the Knesset Vox corpus achieves the best overall performance, significantly reducing errors in complex domains such as Slang and Rare Phonemes compared to both the Phonikud baseline and an ablation trained on legacy data [Morris: don’t understand the ablation]. Best results per row are highlighted in bold. [Morris: error rates should have one decimal place e.g. 49.1 not just 49]

C. Vocalization (Nikud Prediction)

G2P is similar to diacritization which is an important task, but nikud data is not scalable unlike audio. To demonstrate our per-character architecture, we evaluated its capacity to perform standard Hebrew text diacritization (*nikud*). While our primary G2P pipeline scales by leveraging audio supervision, diacritization remains a foundational task in Hebrew natural language processing (NLP) that shares the exact same core structural challenge: resolving highly localized, context-dependent ambiguities at the grapheme level. Following the per-character classification methodology established by DictaBERT [7], [Morris: need better explanation - we’re using the same methodology as dicta-bert, cite them] we retained our pre-trained ReNikud encoder but replaced the phonetic classification heads with a head tailored to directly predict the correct *nikud* class.

This adapted model was fine-tuned on a dataset of 5 million unvocalized sentences sampled from the Knesset corpus that was used in phonikud training but without the stress, shva and the prefix tags. Phonikud’s data was generated from Dicta Nakdan API. from the 5m sentences we sampled 10k, 25k, 50k and 100k sentences and other 10k sentence for validation. For evaluation we took 100 sentences from nakdimon test and manually fixed the 100 sentences.[Max: how do I need to explain the why its only 100 sentences? its so small because we just want to prove a point no?] [Morris: what does that mean? llm speak]. To generate the ground-truth diacritics for this subset, we utilized the Dicta Nakdan API[Max: true but we just used the phonikud data without stress, shva and the prefix tags] [Morris: why? and how is it the Phonikud corpus? where do the GT diacritics come from? is this just distilling dicta-bert?].

We

The comparative results against the state-of-the-art DictaBERT-Menaked⁴ model are detailed in Table VI. Despite the limited fine-tuning data [Morris: already said that], our adapted model achieved an identical Exact Match (EM) score of 0.29 on the test set. At the corpus level, ReNikud maintained a Diacritic Error Rate (DER) of 2.07%, closely tracking DictaBERT-Menaked’s 1.65%. This confirms

Model	WER (Corpus)	CER (Corpus)	DER (Sample)	DER (Corpus)	EM
DictaBERT-Menaked	10.21	1.68	1.51	1.65	0.29
ReNikud^V (Ours)	12.34	2.11	1.88	2.07	0.29

Table VI: Diacritization performance on the 100 manually corrected Nakdimon benchmark sentences. Metrics are presented as percentages for Word Error Rate (WER), Character Error Rate (CER), Diacritic Error Rate (DER), and Exact Match (EM). Despite training on a fraction of the data [Morris: how much?], the adapted ReNikud (ReNikud^V – ReNikud adapted for vocalization) matches the Exact Match baseline. [Morris: not so convincing - other metrics are still worse; can we compare to fine-tuning DictaBERT directly on the same data? only need one decimal place for percentages e.g. 10.2 not 10.21. Also EM should be percentage with one decimal place]

that our base encoder serves as a robust initialization for diacritization tasks, successfully absorbing the distilled knowledge to achieve competitive accuracy. [Morris: not that convincing—it performs worse and was just distilling dicta-bert-menaked]

To illustrate the quality of the model’s outputs, Table VII showcases examples of unvocalized input text successfully mapped to its fully diacritized form by our adapted model.

D. Ablation and Architectural Comparison

To isolate the contributions of our core design choices and validate the efficacy of our approach, we designed evaluations to justify two key components: our novel classification architecture and the necessity of audio-derived weak supervision. The comparative results of these evaluations are detailed in Table VIII.

Architectural Ablation: To demonstrate that our independent classification head is superior for Hebrew G2P, we compared it against two standard architectures trained on the identical weakly-supervised Knesset Vox dataset. The baselines included a sequence-to-sequence approach (ByT5-Small) and a Connectionist Temporal Classification (CTC) network built on the same dicta-il/dictabert-large-char base encoder used in our method.

[Morris: can we show qualitative examples to illustrate that seq2seq has types of hallucinations that we don’t?]

⁴dicta-il/dictabert-large-char-menaked

Unvocalized Input	ReNikud ^V	Dicta
מגיע הרגע בו אנחנו מרגישים שחייבים לעשות שינוי	מגיע הרגע בו אנחנו מרגישים שחייבים לעשות שינוי	מגיע הרגע בו אנחנו מרגישים שחייבים לעשות שינוי
כל ראש ממשלה ואופיו	כל ראש ממשלה ואפיו	כל ראש ממשלה ואפיו
אם אתם מוגדרים כניצולי שואה אנא לחצו כאן	אם אתם מוגדרים כניצולי שואה אנא לחצו כאן	אם אתם מוגדרים כניצולי שואה אנא לחצו כאן

Table VII: Examples of the adapted ReNikud model (ReNikud^V – ReNikud adapted for vocalization) acting as a *nakdan*, demonstrating accurate morphological and contextual diacritization, compared with Dicta. [Morris: Dicta what? DictaBERT-Menaked? Dicta API? Missing GT. Also results are not so convincing, it would be better to compare to training DictaBERT on same data, and highlighting words with errors in red (readers will not notice otherwise)]

Qualitative analysis of the predictions highlights a critical flaw in the Seq2Seq baseline: autoregressive hallucination. When encountering modern loanwords or slang that use silent vowel letters (*matres lectionis*), the Seq2Seq model frequently hallucinates non-existent syllables and glottal stops (/ʔ/). For example, for the slang כאפה (/k'afa/), ByT5 predicts /kaʔaf'a/; for האקר (/h'akeʁ/), it outputs /haʔ'ekεʁ/; and for דאחקה (/d'aχka/), it generates /daʔaχak'a/. Conversely, our ReNikud architecture structurally prevents these hallucinations. [Max: By enforcing a constrained, 1-to-1 per-character mapping, the classifier seamlessly resolves these silent letters to the null (∅) class, guaranteeing alignment and generating the exact ground truth pronunciations.] [Morris: these are not good examples of hallucinations that we prevent, I thought seq2seq sometimes predicts things that don't match the Hebrew characters?]

Audio Supervision vs. Text Data: To justify our weak supervision pipeline, we compared the ReNikud architecture trained on our new Knesset Vox audio pseudo-labels against the same architecture trained on, text-derived Phonikud dataset

Training Details: All models were continuously evaluated using a validation set of 250 gold sentences. [Max: what is expected from this part?]

Ablation Results: The results in Table VIII confirm the independent contributions of both our direct phonetic classification architecture and our weak-supervision data pipeline. Structurally, our novel IPA triplet classifier achieves the lowest error rates across all splits (26.68% Test WER / 8.48% Test CER), outperforming alternative generation models trained on the exact same Vox dataset, namely the Seq2Seq baseline (32.12% Test WER) and the CTC network (27.86% Test WER). This performance gap is most pronounced on the curated validation set of complex edge-cases, where our IPA classifier drops WER to 13.71%, compared to 21.18% for the CTC approach and 24.10% for Seq2Seq. [Morris: don't need to repeat all numbers from tables, text is for high-level analysis]

Furthermore, the dataset ablation isolates the specific impact of the audio supervision. When the exact same ReNikud IPA

Method	Validation Set (250)		Test Set (1,500)	
	WER	CER	WER	CER
Seq2Seq	24.10	4.94	32.12	11.00
ReNikud (Phonikud Data)	23.36	6.78	32.59	10.98
CTC Loss Network	21.18	3.76	27.86	8.86
New Classifier Head (Ours)	13.71	2.69	26.68	8.48

Table VIII: Performance comparison of generation methods. Error rates are reported as percentages for Word Error Rate (WER) and Character Error Rate (CER), evaluated on both a curated validation set of 250 sentences and the full held-out test set of 1,500 sentences. [Morris: only need one decimal place in results e.g. 24.1 not 24.10] [Morris: maybe remove val set]

classifier architecture is trained solely on the legacy text-derived Phonikud dataset, Test WER degrades from 26.68% to 32.59%. This confirms that while reframing G2P as a direct IPA classification task structurally prevents generative hallucinations, the architecture alone is insufficient.

[Max: im not sure how to add the gemini results]

[Morris: these are confusing and need more explanation]

IV. RELATED WORK

Explicit G2P conversion for Hebrew was recently introduced by Kolani et al. [3], following prior works on Hebrew diacritization [6], [7]. These approaches are bottlenecked by the availability of annotated textual data, while we use audio as a scalable source of pronunciation information.

Among prior works learning pronunciation from audio in other language settings, we distinguish between two approaches:

(1) **Audio-supervised** methods such as ours use audio during training to improve pronunciation knowledge. Most similar to our work, a few studies use audio to improve G2P for English [11], [12] or in multilingual settings [13]. However, these works train on labeled audio as a supplementary signal to labelled text, while we use large-scale unlabeled audio as our primary training signal. Additionally, we operate on the Hebrew language which has a high degree of orthographic ambiguity and phonetic features unspecified in text (stress, spoken norms, etc.), and our pseudo-vocalization architecture avoids seq2seq hallucinations by enforcing abjad-style alignment.

(2) **Audio-guided** methods also take audio along with text at inference time for G2P [14]–[16] or abjad diacritization [17], [18]. While adding audio as an additional input can provide a richer signal, we focus on the case where only text is available at inference time.

V. CONCLUSION

We introduced ReNikud, a *Pseudo-Vocalization* architecture for Hebrew G2P conversion. By reframing the generative sequence-to-sequence problem into a per-character phonetic triplet classification task (Consonant, Vowel, Stress), our method structurally mitigates the generative hallucinations observed in standard Seq2Seq models. [Morris: this paragraph

should be changed, renikud is also a method for using audio, not just the architecture] [Morris: change claim to be about reducing degrees of freedom]

[Yakov: Add a mention about the limitation where, due to our strict alignment, the aligner might drop some informal variants such as חִיטוֹב, which might be pronounced as jixtov.]

To train this architecture, we developed a weak-supervision pipeline utilizing a monotonic FST to align continuous IPA transcriptions from an adapted Whisper ASR model with unvocalized Hebrew graphemes. Our evaluations demonstrate that ReNikud lowers Word Error Rates compared to the legacy Phonikud baseline and the evaluated Seq2Seq and CTC architectures [Morris: phrase that part better], particularly on complex edge-cases such as colloquial slang, rare phonemes, and penultimate stress. Furthermore, we validated the extensibility of our core character-level encoder by successfully adapting it for Hebrew text diacritization via knowledge distillation, using only a fraction of standard training data volumes. [Morris: llm speak]

Limitations and Future Work: A primary limitation of our methodology is its reliance on ASR-generated pseudo-labels, as inherent transcription errors from the ASR model naturally propagate into the G2P training data. Additionally, because the Knesset Vox dataset consists exclusively of parliamentary speeches, the training corpus carries a distinct formal and demographic bias. [Morris: LLM speak] This restricts the model’s exposure to diverse, casual, and second-person conversational phonology—a domain shift that accounts for the observed performance gap in resolving gendered morphology compared to legacy models trained on broader literary text. [Morris: are you sure that’s true?] Future work will focus on expanding our weak-supervision pipeline to incorporate more diverse, unstructured conversational audio corpora to address these domain-specific biases. Furthermore, because the core challenge of unwritten vowels is shared across other abjad writing systems (such as Arabic), we plan to extend the *Pseudo-Vocalization* architecture to these languages to evaluate its cross-lingual generalizability. [Max: something do to with חִיטוֹב וְנִיבּוֹת] [Morris: what do they limit?]

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